Task 5 Report

Phase 1: Model Preparation & Fine-Tuning

A ResNet-34 model was fine-tuned on the Caltech-101 dataset using transfer learning to establish a strong baseline accuracy. All pre-trained convolutional layers except the final block were frozen, and only the final fully-connected layer was trained to classify the new dataset classes.

Phase 2: Adversarial Attack Implementation

The Fast Gradient Sign Method (FGSM) was implemented to generate adversarial perturbations that successfully fool the model into making incorrect predictions. The attack's effectiveness was evaluated by varying the epsilon parameter to determine the minimum perturbation required for consistent misclassification.

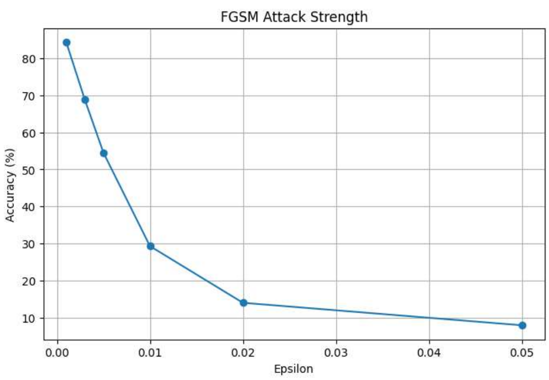
Phase 3: Model Explainability Implementation

Vanilla Saliency Maps and Grad-CAM were implemented to visualize the internal decision-making process of the CNN. These techniques generate heatmaps that highlight the exact pixels the model relies on when making its classifications.

Phase 4: Forensic Analysis

The model's vulnerability began by varying the attack strength (epsilon) of the FGSM. As illustrated in the FGSM Attack Strength graph, the model's accuracy degrades rapidly as the perturbation magnitude increases.

- Low Perturbation (ϵ=0.001): maintain an accuracy of approximately 84%
- Critical Threshold (ϵ=0.01): the model's accuracy collapses to 29.32%. This indicates that small changes to the pixel values are enough to break the model's decision boundaries
- High Perturbation (ϵ=0.05): The model is effectively blinded, with accuracy dropping to single digits (8%)



Phase 5: The Countermeasure Protocol (Model Hardening)

Adversarial training was implemented to harden the model by training it on a combination of clean and adversarially generated examples. The defense's effectiveness was quantitatively proven by comparing the original and hardened models, demonstrating a significant increase in accuracy against adversarial attacks while maintaining high clean accuracy.

Model State	Clean Accuarcy	Adv Accuarcy (e = 0.01)	Performance drop
Original	91.33%	29.32%	-62.01%
Hardened	94.70%	73.60%	-21.10%